A new and Reliable Decision Tree Based Small-Signal Behavioral Modeling of GaN HEMT

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***Abstract*—The paper, for the first time, explores multivariable small signal modeling technique of GaN HEMT based on Decision tree. The proposed model presents a novel binary decision tree to model the GaN HEMT device for multi-biasing and broad frequency range. Bayesian algorithm has been used to find the optimal hyperparameters for better generalization capability and higher accuracy. An excellent agreement is found between the measured S-parameters and the proposed model for complete frequency range of 1GHz-18GHz.**

***Keywords*—GaN HEMT, Small-signal model, Decision Trees, CART**

I. Introduction

RF power amplifier is the last component of transmitter chain before transmission of signal to antenna. Currently, Gallium Nitride (GaN) is considered the most promising device technology for high power amplifier design owing to the exciting material properties of GaN. The advanced traits of GaN such as wide bandgap (3.4 eV), high breakdown voltage, high electron mobility (1500 cm^2 V^{-1} s^{-1}) and high frequency operation (device geometry in micrometer), and ability to form heterojunctions (High Electron Mobility Transistor) make it an ideal candidate for designing of high-power amplifiers (HPAs)[1] – [2]. However, GaN device is still not widely used in the design of HPAs due to unavailability of generic large signal model. An accurate small signal model is the primary requirement in building an overall model, before realization in CAD tool for the analysis and simulation of high frequency circuits and systems.

Machine Learning technique is one of the powerful and contrary technique to conventional methods [3] – [5], capable of handling multi-dimensional and multivariable data in dynamic environments that aids in modeling the device behavior without the knowledge of device physics, expedites the design optimization process and avoids the solution of numerous equations. Artificial Neural Networks and Support Vector Regression have been earlier used to model the GaN HEMT device [6] – [10]. However, other ML techniques have not been employed in device model development. It is, therefore, imperative to explore other advanced ML techniques in model development to assess their corresponding effectiveness.

This paper develops a GaN HEMT modeling technique based on Decision Tree (DT), a machine learning based classification and regression tool, having a tree like structure, where each internal node denotes a test on attribute, each branch represents an outcome of test, and each leaf node (terminal node) holds class label. DT is a supervised machine learning based technique and is used to develop model the small signal behavior of GaN HEMT device. Decision tree technique possess an asset such as simple, highly super interpretable, avoids or does little data pre-processing, high generalization capability that is enough to look beyond ANN and SVR. The proposed model effectively and accurately models the GaN HEMT small signal behavior. A comparison has been drawn between the proposed model and the conventional techniques. The proposed model is found to have better generalization capability than the conventional learning techniques such as SVR and ANN. Moreover, the proposed technique requires less training parameters to tune and optimize than SVR and ANN.

The subsequent sections highlight the device characterization, stepwise framework for the development of small signal model, model validation and finally the conclusion.

II. Steps For The Development Decision Tree Based Small-Signal Model of GaN HEMT

A. Basic Structure of Decision Tree

Regression based Decision tree [11] builds model in the form of tree structure as shown in Fig. 1. The decisions in the tree begin from the root node and terminates at leaf node, leaf node containing the responses.

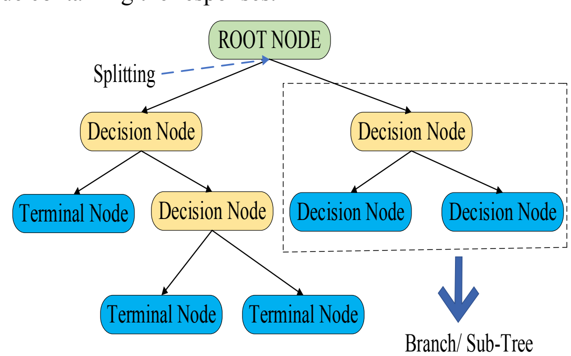

Fig. 1. Basic Structure of Decision Tree.

The trees are fragmented into smaller subsets (known as branch or subtree) and an associated decision tree is simultaneously developed. A decision node two or more branches, each representing values for the tested predictors. The leaf node represents a final response on the target. The topmost decision node in a tree is called the root node that corresponds to the best predictor.

B. Predictors and Responses of the Model

GaN HEMT of device geometry 4x100µm , fabricated device not shown here due to manufacturer policy, as shown in Fig. 2. GaN HEMT is characterized in terms of magnitude and phase of two port S-parameters for multi-biasing set of V_{GS} (gate to source voltage), V_{DS} (V) (drain to source voltage) and broad frequency range of 1-18GHz. Here, the source terminal of the device is grounded.

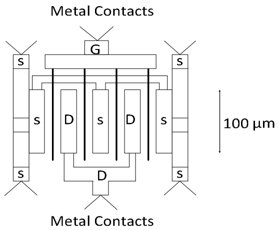

Fig. 2. Layout of 4x100µm GaN HEMT device.

The predictors selected for the proposed model are V_{GS} (V), V_{DS} (V) and frequency (GHz) and the responses are magnitude and phase of two-port S-parameters. However, the real and imaginary parts of S-parameters can also be the responses to the model, S-parameters being a complex number. V_{GS} (V) is varied from [-7V,0V] with a step size of 1V and V_{DS} (V) is varied from [0V, 10V] with a step size of 2V. The objective is to utilize the measured data and build a decision tree based behavioral model of GaN HEMT. The proposed decision tree based small signal model of GaN HEMT is shown in Fig. 3.

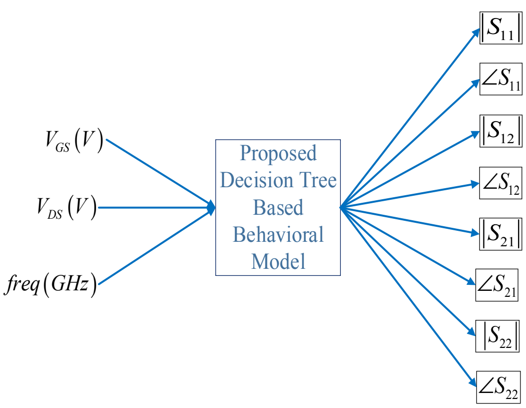

Fig. 3. Decision Tree based Small Signal Model of GaN HEMT.

C. Proposed Small Signal Model of GaN HEMT

The decision tree based behavioral model of GaN HEMT, shown in Fig. 3 has 3 predictor variables and 8 response variables. 67% measured data is utilized for training the data and the remaining 33% is kept reserved for testing and validation. The decision tree based model grows like a tree based structure as shown in Fig. 1. The basic principle of the tree growing process is to select a split among all the splits in a manner that the resulting child node remains the purest. If X is a continuous variable having Q different values, in case of regression, there exist Q-1 different possible splits on X. The recursive procedure of the tree growing process is described in subsequent steps:

Step 1: For all the predictor variables x_i, i=1,2,…..n. Evaluate the mean square error (MSE) of the responses in node q using

$$e_q = \frac{1}{N}\sum_{i\in T}\left(y_i - \overline{y_t}\right)^2 \quad (1)$$

i is the observation and T is the observation indices of node q. The probability that an observation lies in node q using

$$p(T) = \frac{1}{N} \qquad (2)$$

Where N is the sample size.

Step 2: All the predictor variables are sorted. Each element of the sorted predictor is considered as a split point. The observations with missing values are put in the unsplit set (T_u).

Step 3: The observation in the node q splits into left child node (q_l) and right child node (q_r) . The best split among all the split node is chosen that maximizes the reduction in MSE. The reduction in MSE for the current split node, provided that x_i does not contain missing values, is given by

$$\Delta I = p(T)e_q - p(T_l)e_{ql} - p(T_r)e_{qr} \quad (3)$$

where q_l and q_r contain observation indies in the set's T_l and T_r respectively. However, for the observations randomly missing of the predictor variables, reduction in MSE is calculated as

$$\Delta I_u = p(T - T_u)e_q - p(T_l)e_{ql} - p(T_r)e_{qr} \quad (4)$$

where T_u is set of observation in node q that are not missing.

The algorithm terminates as soon as the maximum reduction in MSE is achieved for a node. The complete process of modeling the GaN HEMT based on decision trees is described in following steps.

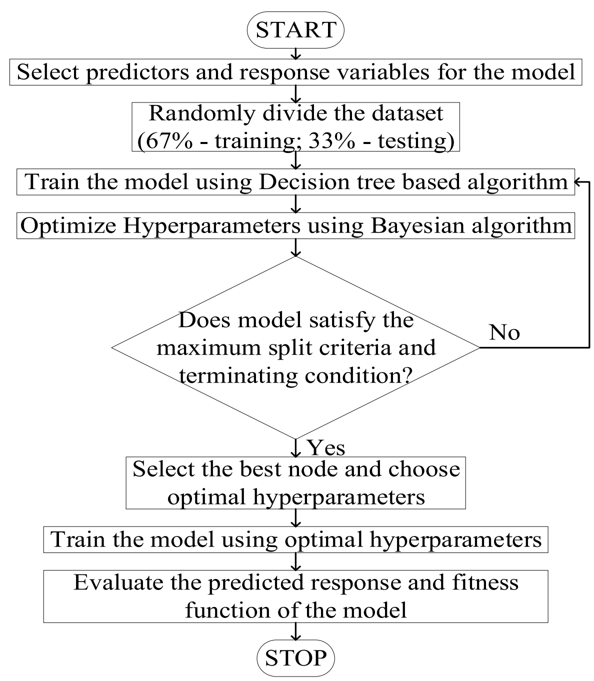

Fig. 4. Flowchart for the Decision Tree based modeling of GaN HEMT.

D. *Results and Discussions*

The optimal hyperparameters that govern the accuracy and generalization capability of the model are maximum number of splits in branch node, minimum leaf size and the minimum parent size. The proposed model splits in a manner that at least one leaf node must be fewer than minimum leaf size and one branch node to be fewer than minimum parent size. The number of children node utilized is determined using the expression Children=2*Number of leaf-1. In our work, the minimum parent size is set to 10. A Bayesian algorithm [12] is used to find the optimal hyperparameters in a manner that must reduce the fitness function and improves the generalization capability of the model. The fitness function of the model is evaluated in terms of Mean Square Error and is given by the expression:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(y_{meas} - y_{pred}\right)^2 \quad (5)$$

where N is the total sample size, y_{meas} is the true value of response variable and y_{pred} is the predicted response. The optimal hyperparameters of the proposed model are put in Table I.

TABLE I. OPTIMAL TRAINING HYPERPARAMETERS OF THE PROPOSED MODEL

Hyperparameters/ Response	**Min. Leaf Size**	**Max. Number of Splits**	**Number of Leaf**
$\lvert S_{11}\rvert$	2	819	726
$\angle S_{11}$	3	1034	325
$\lvert S_{12}\rvert$	2	2076	447
$\angle S_{12}$	3	964	681
$\lvert S_{21}\rvert$	1	591	526
$\angle S_{21}$	1	1477	726
$\lvert S_{22}\rvert$	1	4114	639
$\angle S_{22}$	1	3162	379

The proposed model is trained using the values of optimal hyperparameters and subsequently the predicted responses are evaluated at different multi-bias set for a complete frequency range. The fitness function of the proposed model is evaluated for the training set and the testing and validation set respectively in Table II and Table III.

TABLE II. PREDICTED MSE OF THE PROPOSED MODEL (TRAINING SET)

Bias/ Response	**V_{GS}=0V; V_{DS}=2V**	**V_{GS}=-2V; V_{DS}=8V**	**V_{GS}=-4V; V_{DS}=0V**	**V_{GS}=-6V; V_{DS}=6V**
$\lvert S_{11}\rvert$	1.944e-5	1.784e-5	1.989e-5	1.075e-5
$\angle S_{11}$	6.919e-3	8.217e-4	9.999e-4	5.855e-4
$\lvert S_{12}\rvert$	1.812e-5	1.689e-2	3.286e-5	8.518e-4
$\angle S_{12}$	1.571e-3	4.591e-4	2.457e-4	2.078e-4
$\lvert S_{21}\rvert$	2.461e-6	4.239e-6	3.408e-6	2.972e-6
$\angle S_{21}$	1.922e-4	1.931e-4	1.528e-4	7.400e-5
$\lvert S_{22}\rvert$	1.479e-4	4.954e-5	5.911e-5	4.792e-5
$\angle S_{22}$	2.143e-4	1.272e-3	1.063e-4	5.161e-4

TABLE III. PREDICTED MSE OF THE PROPOSED MODEL (TESTING AND VALIDATION SET)

Bias/ Response	**V_{GS}=-1V; V_{DS}=0V**	**V_{GS}=-3V; V_{DS}=10V**	**V_{GS}=-5V; V_{DS}=6V**	**V_{GS}=-6V; V_{DS}=10V**
$\lvert S_{11}\rvert$	2.107e-5	2.425e-5	6.151e-6	1.360e-5
$\angle S_{11}$	6.882e-3	2.875e-3	5.892e-4	7.433e-4
$\lvert S_{12}\rvert$	1.881e-5	6.109e-2	1.228e-2	1.881e-3
$\angle S_{12}$	1.481e-4	1.172e-3	1.684e-4	1.856e-3
$\lvert S_{21}\rvert$	3.292e-5	2.962e-5	6.182e-6	2.185e-5
$\angle S_{21}$	9.622e-5	7.279e-4	1.547e-4	8.364e-5
$\lvert S_{22}\rvert$	5.971e-5	1.598e-3	2.217e-3	2.427e-4
$\angle S_{22}$	1.181e-4	8.641e-3	1.055e-2	3.612e-4

It is apparent from Table II and Table III that the training predicted error and testing predicted error is almost same. This means that the proposed model possesses great accuracy and generalization capability in terms of magnitude and phase of two port S-parameters.

III. MODEL VALIDATION

The previous section layout the framework for the development of small-signal behavioral model of GaN HEMT based on Decision Tree. This section investigates the reliability of the above procedure. The measured S-parameters were compared with proposed model on test and validation set as shown in Fig. 5.

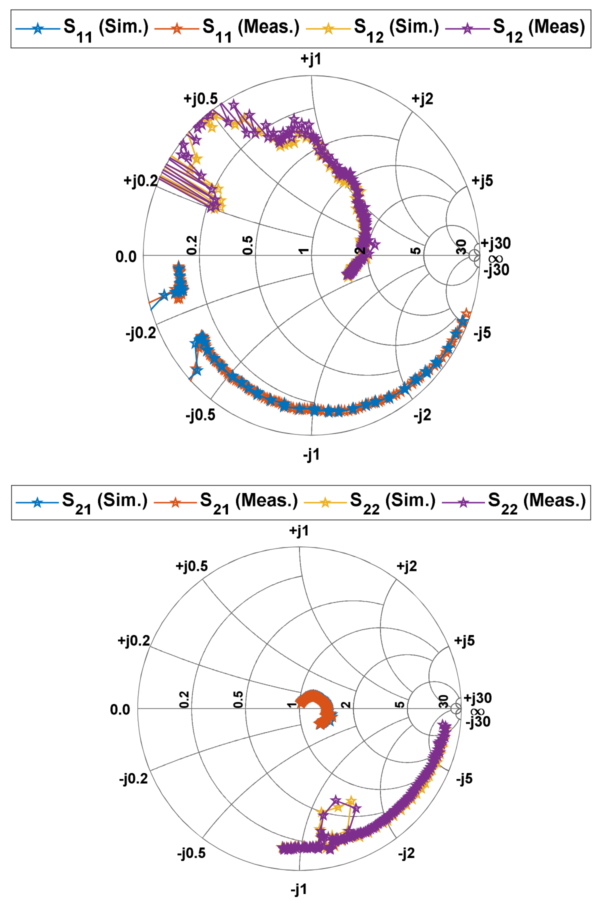

Fig. 5. Comparison between measured S-parameters and the proposed model at V_{GS}=-6V, V_{DS}=10V.

An excellent agreement has been found between the measured S-parameters and the proposed model for a complete frequency range of 1GHz-18GHz. Fig. 5 demonstrates the accuracy of model with the fact that even the sharp transitions observed in S_{12} and S_{22} are accurately modeled.

The proposed technique is compared with the other conventional techniques such as SVR and ANN in terms of the accuracy and generalization capability in Table IV. The accuracy is determined by calculating MSE for complete set of training data and test data. SVR is trained using the Sequential Minimization Algorithm (SMO) and ANN is trained utilizing Multi-Layer Perceptron architecture with 15 neurons in a hidden layer and tan-sigmoid as an activation function. In both the conventional techniques, Bayesian regularization algorithm is used find the optimal parameters, weights and bias in ANN and Penalty Factor and epsilon insensitive band of the kernel function in SVR. It is shown in Table IV that the proposed technique possesses higher accuracy than both SVR and ANN. The proposed technique offers a high generalization capability than conventional ML techniques. It is quite apparent from Table II, Table III, Table IV that training error and testing error is approximately within range of each other.

TABLE IV. COMPARISON BETWEEN THE PROPOSED MODEL, SVR AND ANN.

Methods/ Response	Training Set			Testing Set		
	SVR	**ANN**	**DT**	**SVR**	**ANN**	**DT**
$\|S_{11}\|$	7.1e-3	1.1e-4	1.3e-4	8.2e-3	1.1e-3	6.3e-4
$\angle S_{11}$	5.9e-1	2.9e-1	4.5e-2	1.314	1.000	6.2e-2
$\|S_{12}\|$	6.1e-3	5.2e-3	3.4e-3	1.2e-1	7.9e-2	6.5e-2
$\angle S_{12}$	7.8e-4	8.6e-4	4.5e-4	1.5e-2	4.6e-2	1.4e-3
$\|S_{21}\|$	1.4e-5	1.3e-5	4.6e-6	1.7e-3	1.5e-3	7.8e-4
$\angle S_{21}$	1.2e-3	4.7e-4	1.7e-4	1.7e-2	2.4e-2	1.2e-3
$\|S_{22}\|$	1.1e-3	1.7e-4	1.2e-4	1.1e-2	3.9e-3	2.7e-3
$\angle S_{22}$	7.6e-2	2.7e-2	7.2e-3	4.308	1.954	0.712

IV. CONCLUSION

The paper presents a novel decision tree based small signal behavioral model of GaN HEMT. The proposed model is highly accurate and possess high generalization capability than SVR and ANN for broad frequency range and multi-bias condition. An excellent agreement has been found between the measured S-parameters and the proposed model. The machine learning model is fast and can be embedded into CAD tool for the analysis and simulation of high frequency circuits and systems.

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